MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

Name: Samprita Baidya Class: SOC-09 Batch: B
Subject: PL Python Roll No.: 47 Exp. No.: 1
Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

Q.1) key features of python.

1. Simple and easy to learn syntax.
2. Interpreted language.
3. High level languge
4. object oriented.
5. Platfarn independent.
6. Large Standing librarg.
7. Support multiple programming paradigms.

Advantage of python

1. Faster program development.
2. Easy developing and maintenance.
3. Used in web development, AI, data Science automation.
4. Large community Support.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Python IDEs/Editors Ed
- Pycharm
- Visual Studio Code. (VS code)

Q2
→ History & Evolution
- Python was created by Guido van Rossum in 1989
- Released officially in 1991
- Named after Monty Python
- Python 2 released in 2000
- Python 3 released in 2008 (currently widely used)

→ Why python is popular
- Easy for beginners
- Readable and clean syntax
- Powerful libraries (NumPy, Pandas, Tensorflow)
- Used in industry: Google, Netflix, NASA

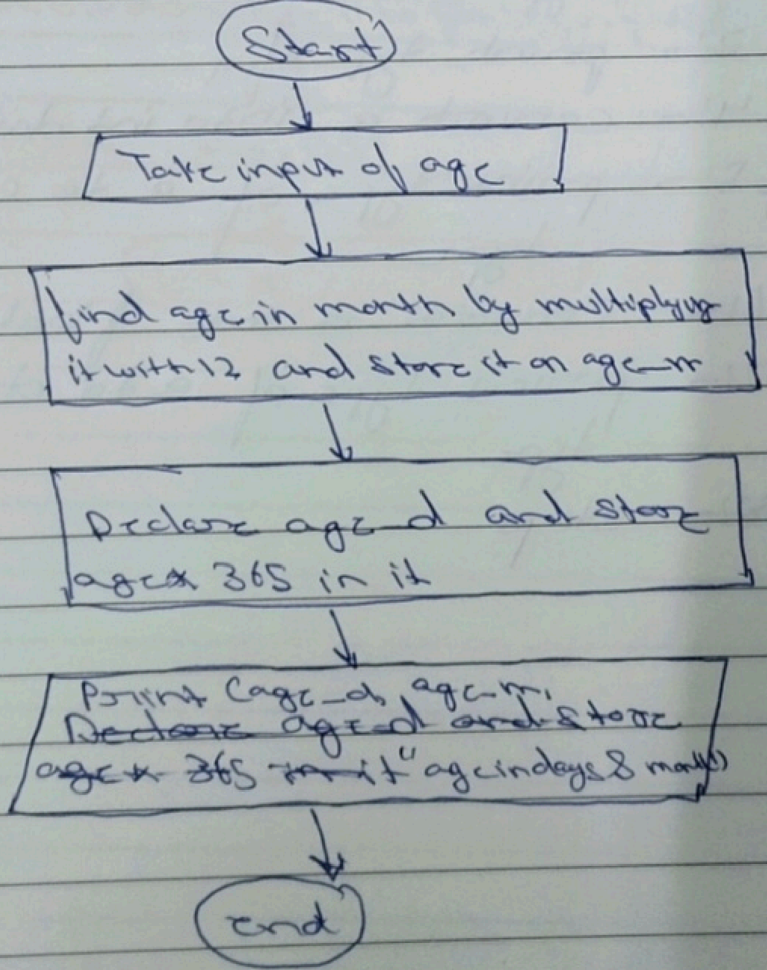
Q3
→
```
#Accept two numbers
a= float(input("Enter first number."))
b = float(input("Enter second number: "))

#Operations
print ("Addition:", a+b)
print ("Subtraction", a-b)
print (" Multiplication:", a*b)
```

Step 3:- Now find age in month by multiplying by 12 and store it on variable age_m

Step 4:- Now find age in ~~month~~ days by multiplying by 365 and Store it on variable age_d

Step 5:- Print the age_m and age_d
Step 6:- End

Flow Chart:-



print f("Division:", a/b)

Q4
→ a = 10
b = 5.5
c = 3+4j
d = True
print (a+b, type (c), d)

Data Types

variable	value	Data Type
a	10	int
b	85.5	float
c	3+4j	complex
d	True	bool

output
15.5 <class 'complex'> True

Explanation
- a+b → int + float → float result
- Type (c) return complex
- d prints boolean value

Q5
→ Relational Operators
used for comparison

a = 10
b = 5
print (a>b)
used in decision making

Logical Operators
used to combine conditions
print (a>b and b>0)
used in conditions and loops

Bitwise Operators
Used at binary level
x = 5
y = 3
print (x&y)
Used in low-level programming

Q6
→ a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("1. Addition")
print("2. Greater than")
print("3. Logical AND")
print("4. Assignment")

choice = int(input("Enter choice: "))

```
If choice ==1:
    print ("Sum =", a+b)
elif choice ==2:
    print (a>b)
elif choice ==3:
    print (a>0 and b>0)
elif choice ==4:
    a +=b
    print ("updated value:", a)
else:
    print ("Invalid choice")
```

Q7

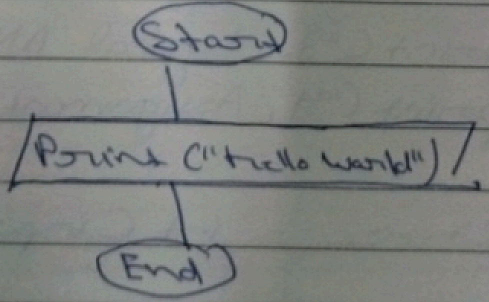
Print ("Hello World")

Algorithm

Step 1:- Start
Step 2:- write print function and write "Hello World" in the print function

Step 3:- Stop/End

flowchart :-



```
If choice == 1:
    print ("Sum =", a+b)
elif choice == 2:
    print (a>b)
elif choice == 3:
    print (a> 0 and b>0)
elif choice == 4:
    a += b
    print ("Updated value:", a)
else:
    print ("Invalid choice")
```

Q7

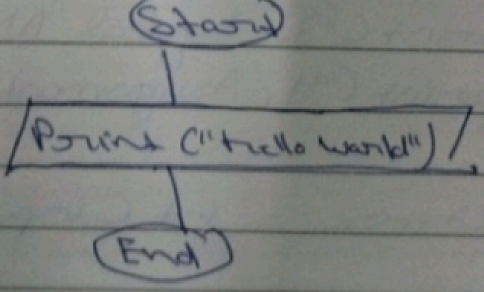
Print ("Hello World")

Algorithm

Step 1:- Start
Step 2:- write print function and write "Hello world" in the print function

Step 3:- Stop / End

flow chart :-



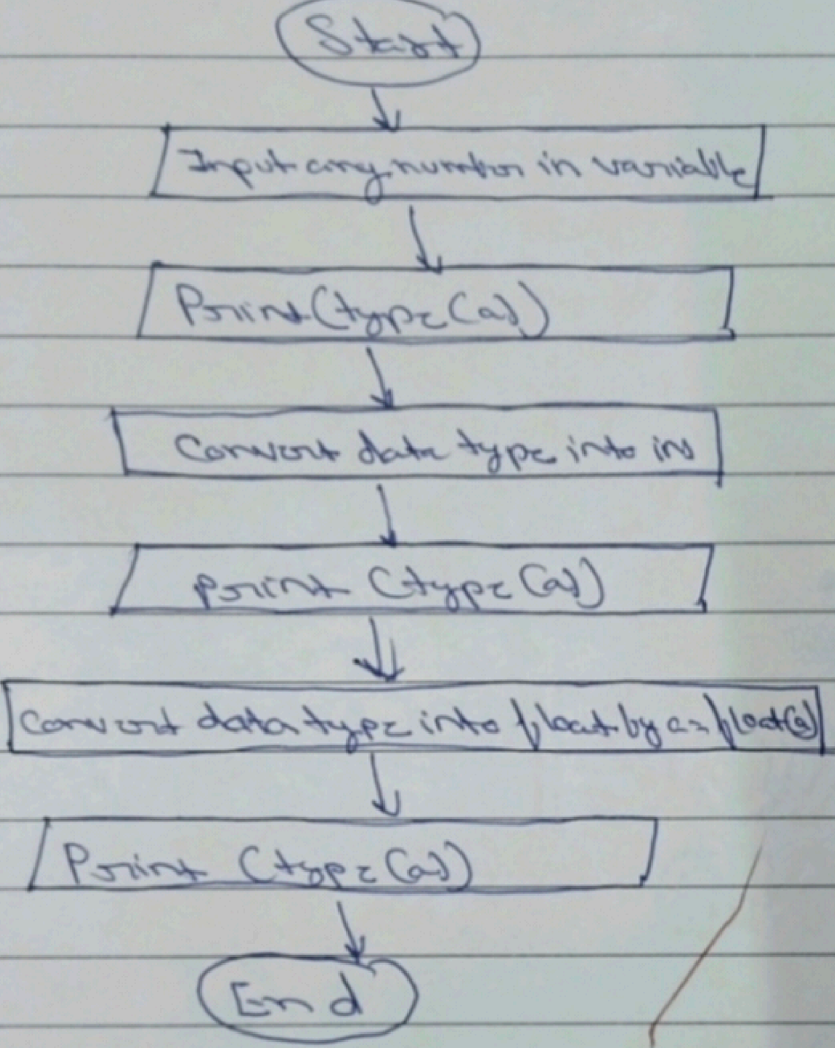
Q8

```
a = input ("Enter first Number")
b = input ("Enter Second Number")
c = input ("Enter third Number")
d = input ("Enter fourth Number")
e = input ("Enter fifth Number")
a = float (a)
b = float (b)
c = float (c)
d = float (d)
e = float (e)
avg = (a+b+c+d+e) / 5
print (". The average of five Number is", avg)
```

Algorithm

Step 1:- Start
Step 2:- Take five input a, b, c, d, e
Step 3:- convert type to float of all five inputs
Step 4:- Add all the 5 inputs and ~~avarat~~ divide by 5 and Store in variable avg
Step 5:- print the avg variable
Step 6:- End

Flowchart:-


Start
Input any number in variable
Print(type(a))
Convert data type into int
print (type(a))
Convert data type into float by a=float(a)
Print (type(a))
End